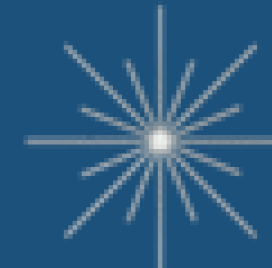


# THE OVER-DESTABILIZATION HYPOTHESIS: BASELINE NETWORK INSTABILITY PREDICTS FAILURE IN HIGH-DOSE KETAMINE PE THERAPY FOR PTSD

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## THE LIMITS OF PLASTICITY: PRE-TREATMENT SN-DMN INSTABILITY STRONGLY PREDICTS CLINICAL FAILURE IN HIGH-DOSE KETAMINE PE THERAPY, REVEALING A CRITICAL STATE-DEPENDENT BOUNDARY CONDITION FOR RECOVERY.

### INTRODUCTION

Post-Traumatic Stress Disorder (PTSD) is increasingly conceptualized as a profound dysregulation of large-scale brain networks [1, 2]. A core pathological feature is the loss of functional segregation between the **Salience Network (SN)** - which mediates external threat detection - and the **Default Mode Network (DMN)**, which supports safety contextualization and self-referential processing [3].

Trauma-focused psychotherapies, such as Prolonged Exposure (PE), rely on fear extinction learning - a process that necessitates robust functional integrity of these fronto-limbic circuits. Subanesthetic ketamine, a rapid neuroplasticity enhancer, is hypothesized to act as a "neuroplastic lubricant" to override pathological deficits and facilitate extinction [4].

However, the efficacy of this pharmacological rescue may be fundamentally **state-dependent**. Recent advances in chronnectomics (dynamic functional connectivity) demonstrate that temporal network instability (dFC variance) is a sensitive index of an erratic, "trauma-locked" brain state [5, 6].

**The Current Study:** We investigated whether high baseline SN-DMN temporal instability acts as a definitive prognostic barrier to recovery during an intensive 1-week PE protocol, and crucially, whether high-dose ketamine could override this "neural noise" or if baseline instability remains an impenetrable boundary condition for therapy success.

### FMRI PROCEDURE & PROTOCOL

Thirty participants (N=30) with chronic PTSD (baseline CAPS-5  $\geq 30$ ) were enrolled in a double-blind, randomized controlled trial. Subjects were randomized into three treatment arms: **High-Dose Ketamine** (0.5 mg/kg; n=11), **Low-Dose Ketamine** (0.2 mg/kg; n=11), or an active **Midazolam Placebo** (0.045 mg/kg; n=8).

**Primary Outcome Measure:** Clinical efficacy was quantified by the absolute drop in CAPS-5 scores from baseline to the 1-month follow-up, where higher positive values indicate greater symptom improvement.

#### Intensive 1-Week Clinical Timeline:

- Day 1 (Baseline):** Resting-state fMRI (MRI1) followed by Infusion 1.
- Days 2-5:** Daily 90-minute intensive Prolonged Exposure (PE) psychotherapy sessions.
- Day 3:** Infusion 2 administered mid-week to sustain the pharmacological window of neuroplasticity.
- Day 6 (Post-Treatment):** Second rs-fMRI (MRI2) acquired 24 hours post-treatment to measure network connectivity shifts.
- Month 1 (Follow-up):** Final CAPS-5 clinical interview to assess therapeutic durability.

### METHODS

**Preprocessing:** Raw resting-state data were processed via the **fMRIPrep** pipeline, providing a robust, reproducible workflow that included susceptibility distortion correction, motion alignment, and spatial normalization to MNI152NLin2009cAsym space, minimizing interpolation artifacts.

**Parcellation:** Brain regions were mapped using the **Schaefer 400-node cortical atlas** integrated with the **Tian subcortical atlas**. This high-resolution fusion ensures precise isolation of core cortical hubs (e.g., ventromedial prefrontal cortex) alongside deep limbic structures critical to fear circuitry.

**Dynamic Connectivity (dFC):** To capture transient state transitions, we calculated the temporal variance of the SN-DMN correlation using a **60-second sliding-window** approach. dFC variance - our primary index of network instability - was defined as the standard deviation of Pearson coefficients across the entire scan.

**Denoising:** We rigorously regressed physiological noise using Anatomical Component Correction (aCompCor) derived from CSF and white matter masks. We explicitly omitted Global Signal Regression (GSR) to preserve the authentic, intrinsic dynamic range of the DMN.

### BASELINE NETWORK INSTABILITY

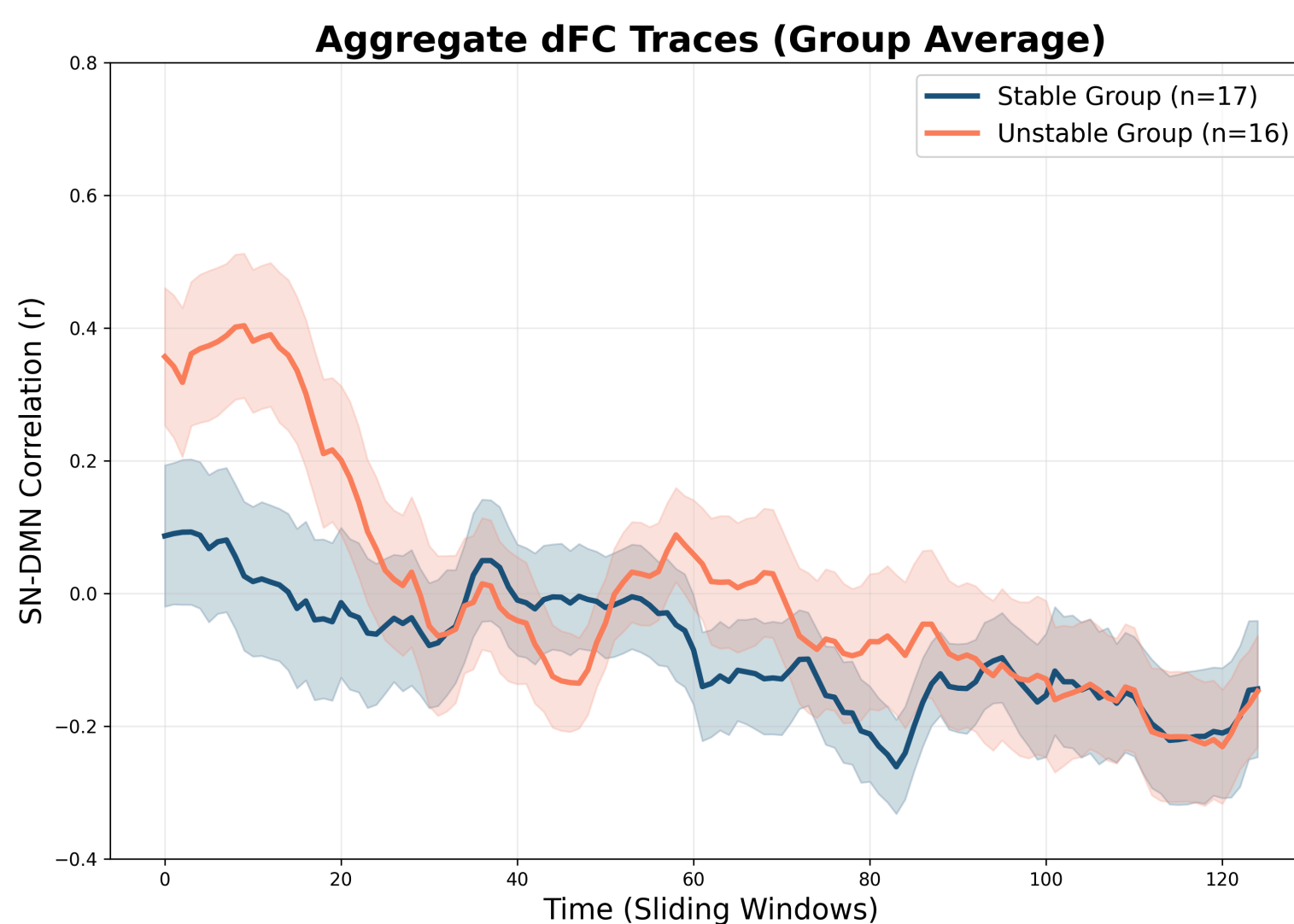


FIGURE 1: Population-Level dFC Profiles. Aggregate SN-DMN connectivity traces (n=33 baseline) split by median instability. (Blue) Stable Group (n=17) shows consistent integration; (Orange) Unstable Group (n=16) shows significant temporal fluctuations. Shaded bands represent standard error.

#### Average Network Instability (Group Baseline)

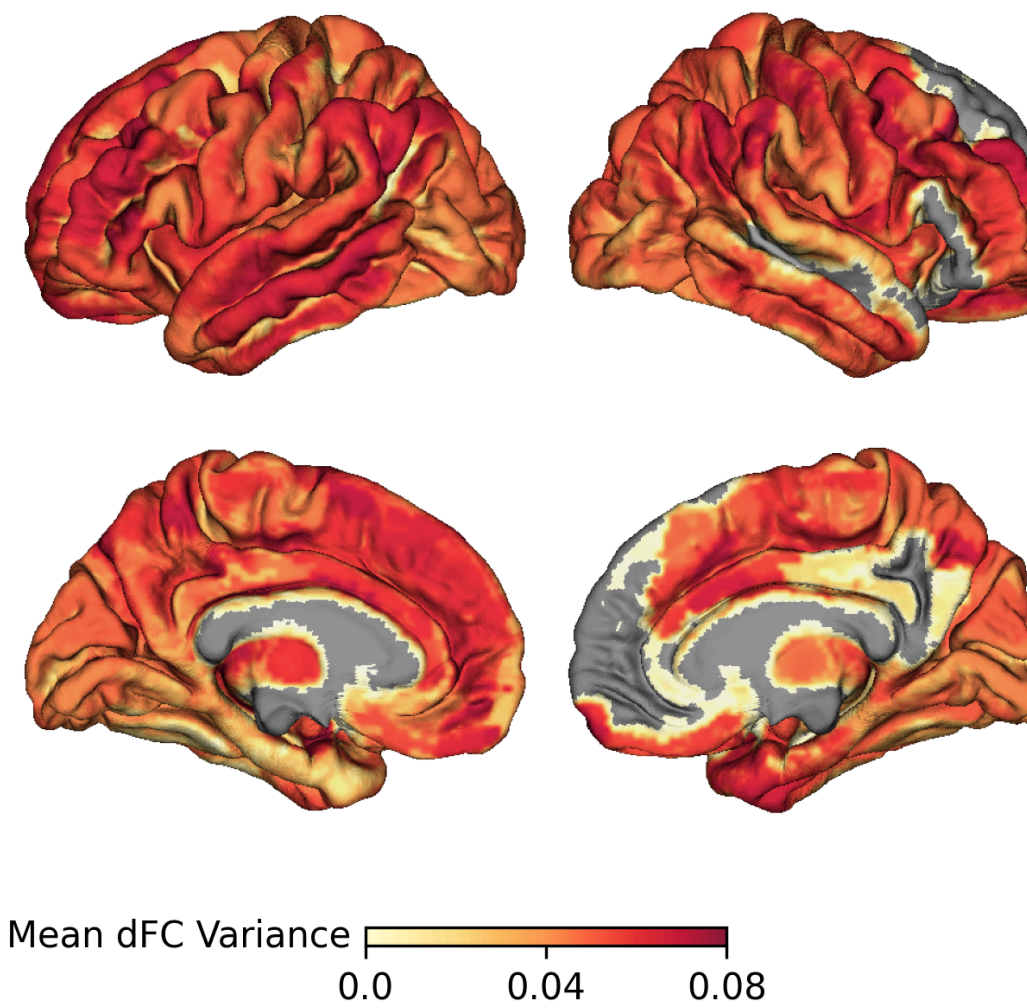


FIGURE 2: Topographic Mapping of Neural Chaos. Group-average dFC variance map (n=33). Hot colors indicate fronto-limbic hubs where connectivity is naturally most dynamic.

This population-level mapping demonstrates that network instability represents a **system-wide dysregulation** of the functional connectome. In individuals with high baseline instability, connectivity fluctuates wildly across fronto-limbic hubs responsible for emotional processing.

Our analysis suggests this chaotic state acts as a definitive **boundary condition**. When the brain is operating near maximum instability, introducing a potent destabilizing agent like High-Dose Ketamine pushes the system past a critical threshold, preventing the consolidation required for successful extinction learning [7, 8].

### PROGNOSTIC RESULTS: BASELINE INSTABILITY PREDICTS FAILURE

Baseline SN-DMN instability served as a robust, dose-specific prognostic biomarker for outcome, validating the **Over-Destabilization Hypothesis**.

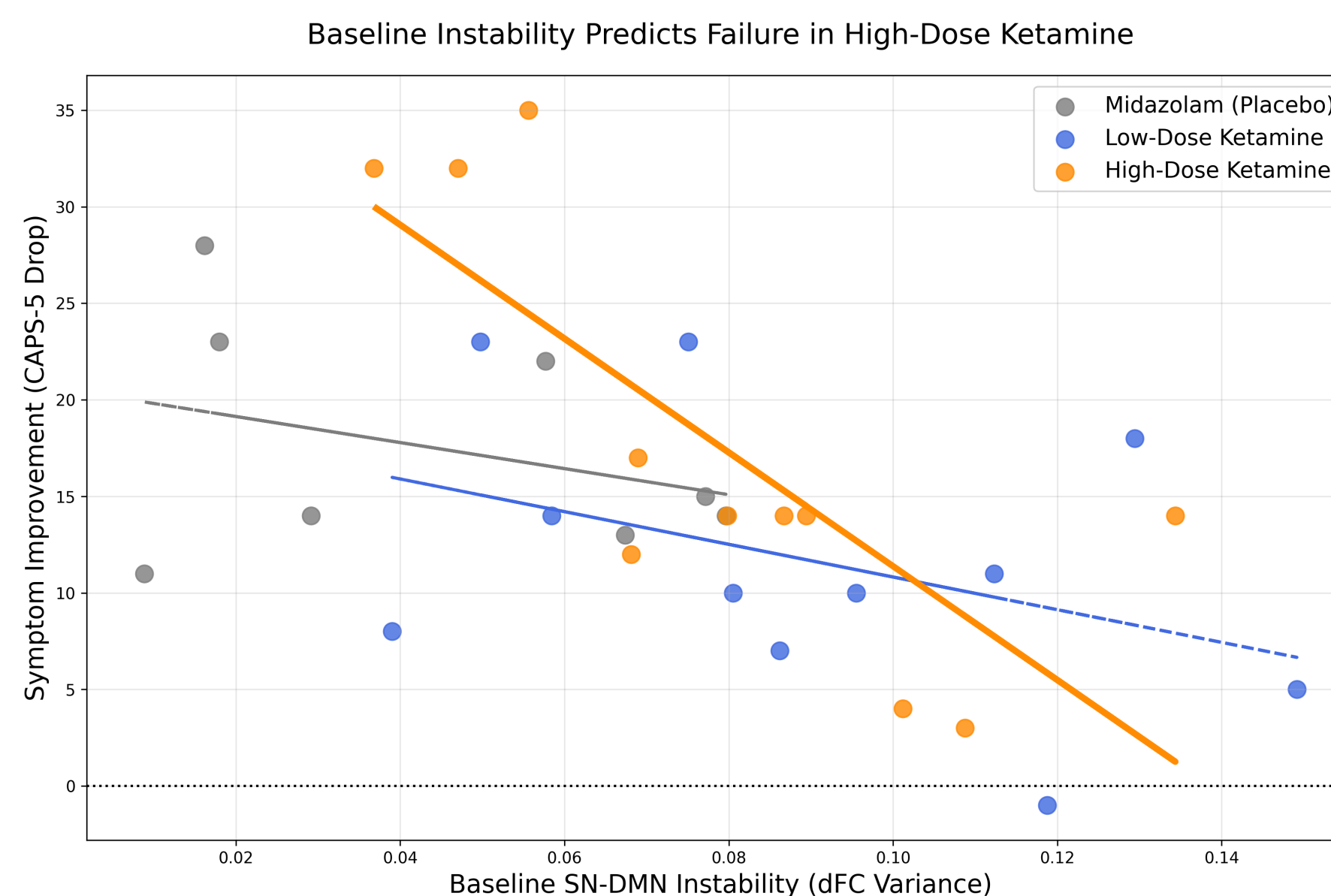


FIGURE 3: In the High-Dose Ketamine group, high baseline instability strongly predicted clinical failure ( $r = -0.770$ ,  $p = 0.005$ ). Placebo and Low-Dose arms did not show this strict dependency.

In the High-Dose Ketamine arm, baseline instability was strongly predictive of clinical outcome, exhibiting a robust negative correlation with symptom improvement ( $r = -0.770$ ,  $p = 0.005$ ). Notably, clinical improvement in the Placebo arm did not exhibit this strict dependency ( $r = -0.328$ ,  $p = 0.42$ ). These distinct correlational trajectories suggest that while an erratic brain state is a core feature of PTSD, **introducing a high-dose pharmacological perturbagen specifically interacts with this instability to create a definitive barrier to recovery.**

### ENTROPY MEDIATION & MECHANISTIC ANALYSIS

To investigate the precise mechanism of recovery, we tested if changes in neural complexity (entropy) acted as a mediator[9]. We hypothesized that Ketamine induces a transient **"Entropy Spike"** at Day 6, creating a window of plasticity[10].

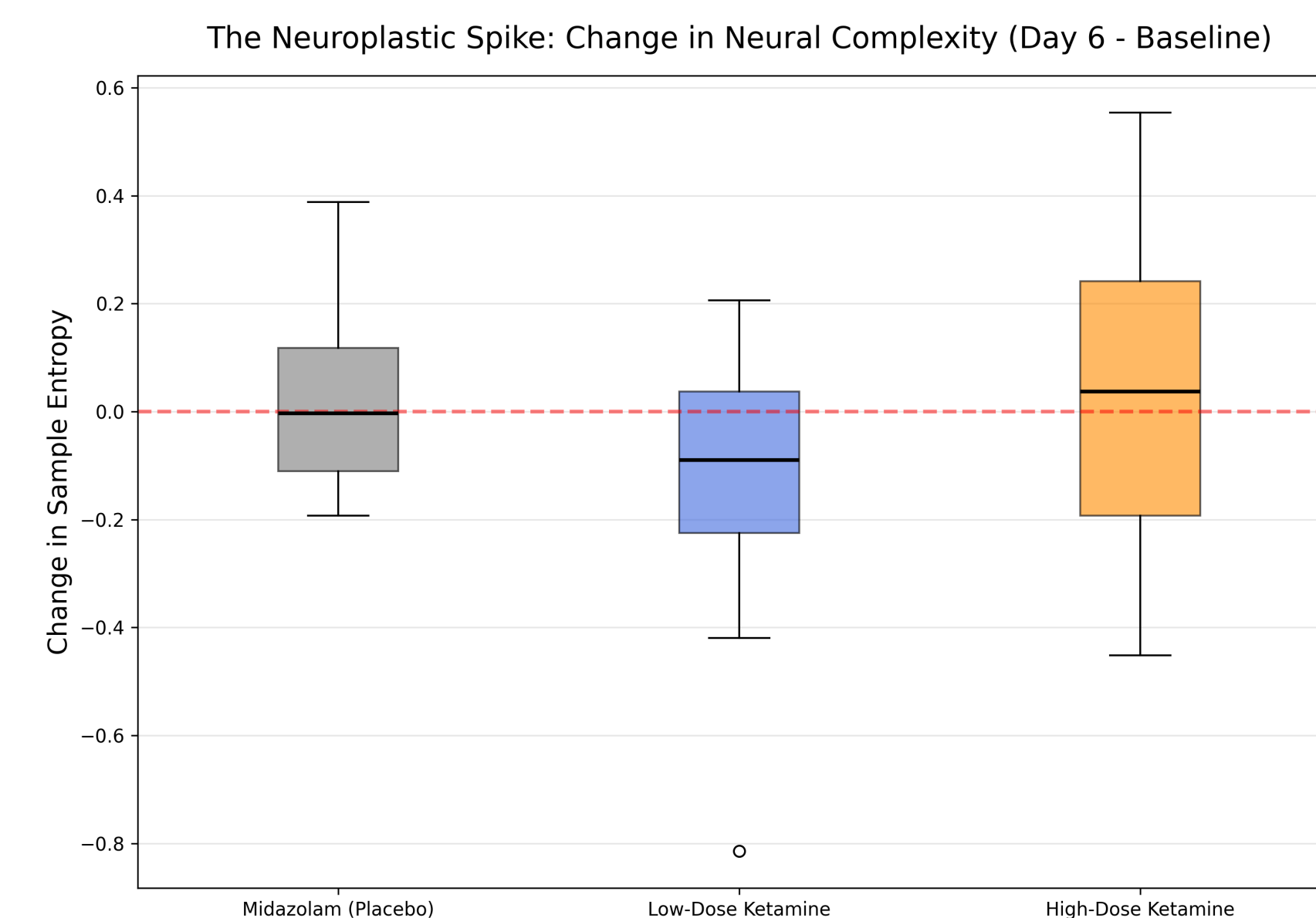


FIGURE 4: Longitudinal analysis revealed that clinical recovery is driven by shifts in neural complexity. While natural variance was beneficial for the Placebo group ( $r = 0.56$ ), the forced 'entropy spike' in the Ketamine group trended toward poorer outcomes.

Overall mediation analysis confirmed that while neural entropy change drives clinical improvement ( $p = 0.04$ ), therapeutic utility remains **state-dependent**. In High-Dose Ketamine patients, the massive entropy spike failed to statistically predict long-term clinical recovery ( $p = 0.324$ ).

### DISCUSSION & CLINICAL CONCLUSIONS

The convergence of our findings robustly supports the **Over-Destabilization Hypothesis** over models of universal neuroplastic rescue. Our results suggest that while therapeutic fluidity is necessary, successful fear extinction requires a stable neural 'scaffold'. For patients within the 'Therapeutic Window' of baseline stability, standard exposure is successful and pharmacological augmentation can optimally enhance consolidation.

However, when the SN-DMN axis is highly unstable at baseline, **the neural architecture becomes too erratic to anchor new safety learning**. In these cases, the potent drug-induced 'entropy spike' appears to push the system past a critical threshold into information-processing chaos rather than constructive plasticity. This creates a definitive boundary condition against pharmacological rescue that ketamine cannot override.

**Clinical Utility:** These results provide a neurobiological framework for personalized triage. By utilizing pre-treatment dFC variance as an objective biomarker, clinicians can identify 'high-instability' phenotypes who require targeted stabilization before attempting intensive pharmacological augmentation, moving the field toward state-dependent PTSD care.

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